Problem 0

(a) Sketch the region $x > y^2 - 1$.

(b) Sketch the region $y > x^2 + 2x + 1$.

(c)* **(hard / bonus / only if we have time)**

Find the area of overlap between these two regions.

Problem 1

(a) Classify each surface as a cylinder, paraboloid, hyperboloid, ellipsoid, sphere, or none of the above.

(i) $z = 2x^2 + 3y^2$ _____

(ii) $x^2 + 4y^2 + 9z^2 = 36$ _____

(iii) $x^2 + y^2 + z^2 - 2x + 4y - 6z = 2$ _____

(iv) $x^2 + y^2 - z^2 = 1$ _____

(v) $z = (x - 1)^2$ _____

(vi) $z^2 = x^2 + y^2$ _____

Problem 1 (continued)

(b) Match each function to its surface below. Each graph is used exactly once.

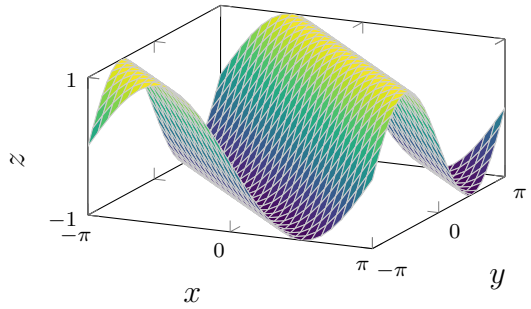
(i) $f(x,y) = \dfrac{1}{1+x^2+y^2}$ (ii) $g(x,y) = 1 - x^2 - y^2$

(iii) $h(x,y) = \dfrac{1}{1+(x^2+y^2-2)^2}$ (iv) $k(x,y) = \cos\left(2\sqrt{x^2+y^2}\right)$

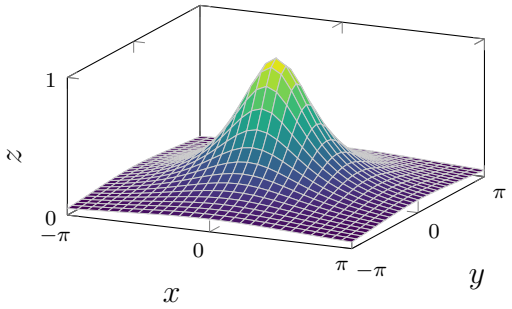
(v) $p(x,y) = \sin x \sin y$ (vi) $q(x,y) = \sin(x+y)$

All plots show $-\pi \le x, y \le \pi$ from the same viewpoint.

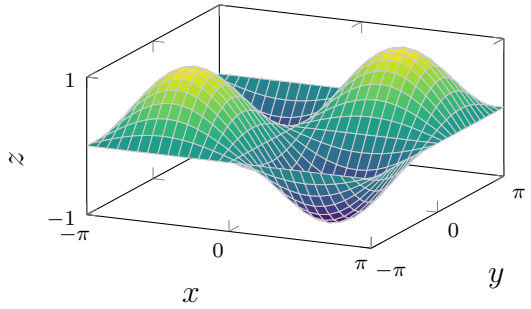
A



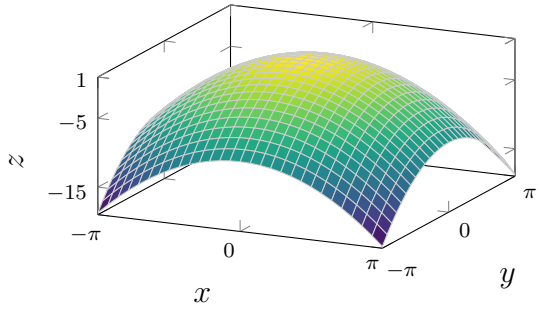
B



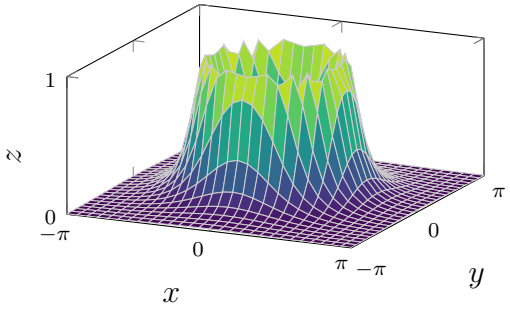
C



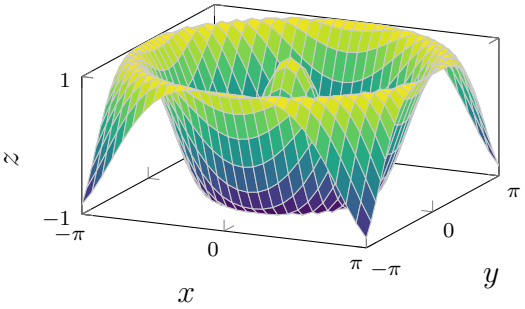
D



E

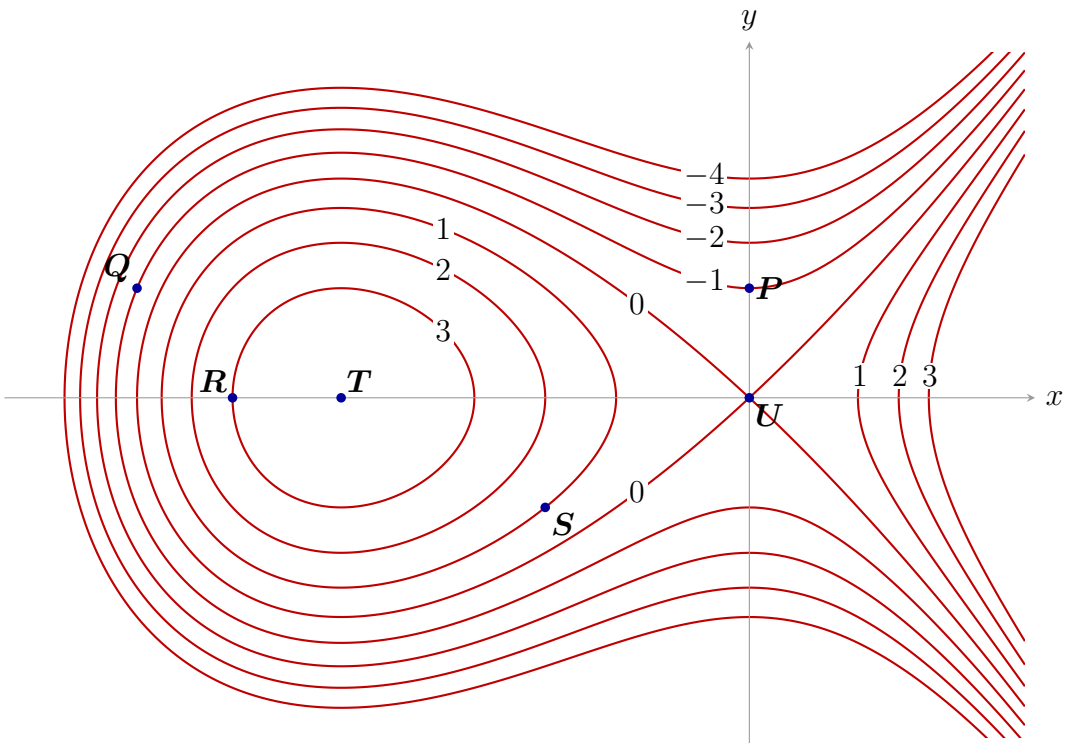


F



Problem 2

The diagram shows level curves of a smooth function $f(x, y)$. For parts (a)–(d), circle all labeled points where the statement applies. Assume that f has a local maximum at T.



(a) $\nabla f = \mathbf{0}$. $P \quad Q \quad R \quad S \quad T \quad U$

(b) f has a saddle point. $P \quad Q \quad R \quad S \quad T \quad U$

(c) $f_y > 0$. $P \quad Q \quad R \quad S \quad T \quad U$

(d) The directional derivative along $\langle 0, -1 \rangle$ is negative. $P \quad Q \quad R \quad S \quad T \quad U$

(e) Draw the gradient at each labeled point. Which point has the largest gradient magnitude?

Problem 3

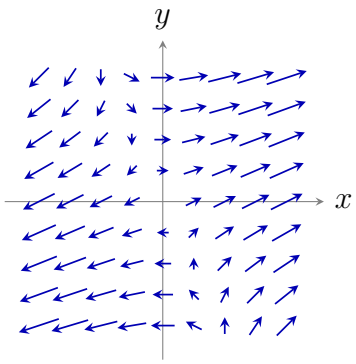
(a) Match each vector field to a plot below.

(i) $\langle -y, x \rangle$ (ii) $\langle x - y^2, y \rangle$ (iii) $\langle x + y, x + y \rangle$ (iv) $\langle y, x \rangle$

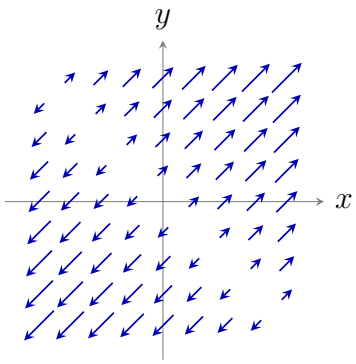
(b) Match each function to the plot of its gradient.

(i) $\frac{1}{2}(x^2 + y^2)$ (ii) xy (iii) $(x + y)^2$ (iv) $x(x + y)$

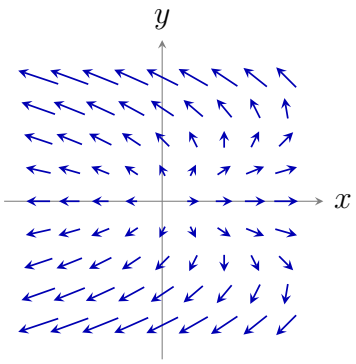
A



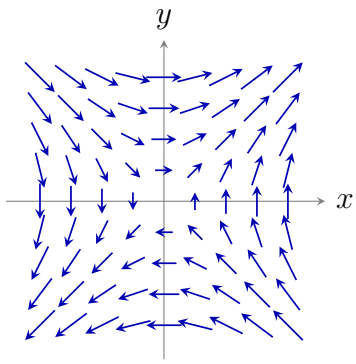
B



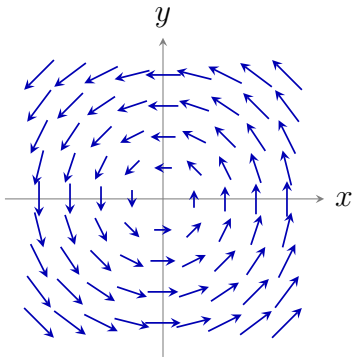
C



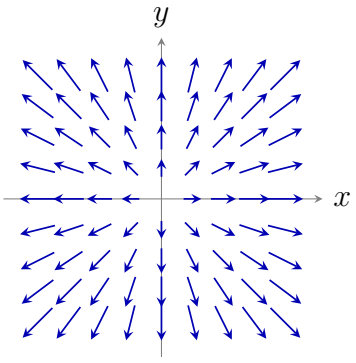
D



E



F



(c) Draw the unit circle, *positively oriented*, on each plot. (If you haven't covered this in class, it means that the interior of the region is on your left as you travel around the curve.) For each field, is its vector line integral around this circle positive, negative, zero, or hard to tell just from looking?

Problem 4

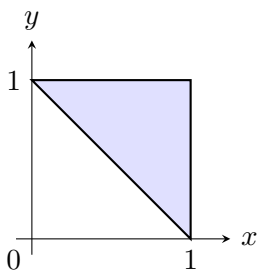
(a) Match each integral to its region below. Do not evaluate.

(i) $\displaystyle\int_0^{\sqrt{2}} \int_y^{\sqrt{4-y^2}} dx\, dy$ (ii) $\displaystyle\int_0^1 \int_{x^2}^x dy\, dx$ (iii) $\displaystyle\int_0^1 \int_{1-y}^1 dx\, dy$

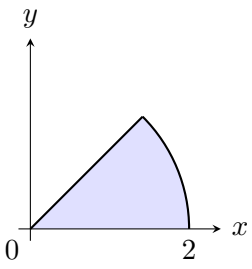
(v) $\displaystyle\int_0^1 \int_{\sqrt[3]{1-y}}^1 dx\, dy$ (vi) $\displaystyle\int_0^1 \int_0^{x^3} dy\, dx$

(iv) $\displaystyle\int_{-2}^{-1} \int_0^{\sqrt{4-x^2}} dy\, dx + \int_{-1}^1 \int_{\sqrt{1-x^2}}^{\sqrt{4-x^2}} dy\, dx + \int_1^2 \int_0^{\sqrt{4-x^2}} dy\, dx.$

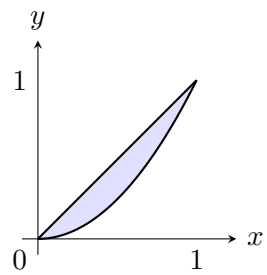
A



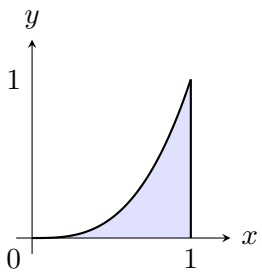
B



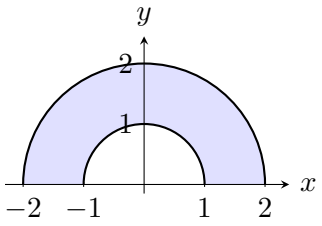
C



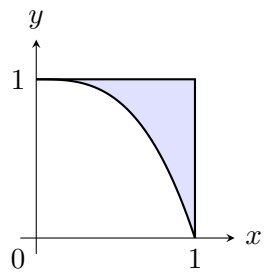
D



E



F

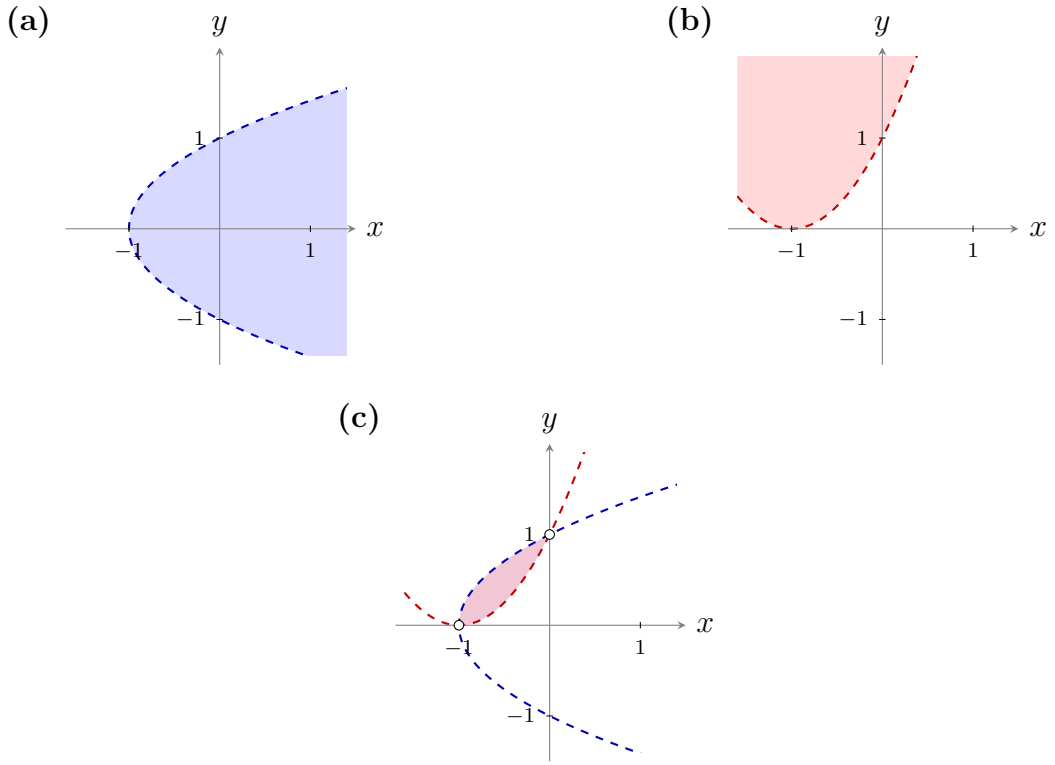


(b) Rewrite each integral in a different (preferably simpler) order. Do not evaluate.

Problem 0: Answer Key

(a) Draw the right-opening parabola $x = y^2 - 1$, with vertex $(-1, 0)$, as a dashed boundary. Shade to its right.

(b) Write $y = (x+1)^2$. Draw this upward-opening parabola with vertex $(-1, 0)$ as a dashed boundary, and shade above it.



(c)* The shaded overlap is bounded above by $y = \sqrt{x+1}$ and below by $y = (x+1)^2$, between $x = -1$ and $x = 0$.

$$\begin{aligned}\text{Area} &= \int_{-1}^{0} \left[\sqrt{x+1} - (x+1)^2\right] dx \\ &= \left[\frac{2}{3}(x+1)^{3/2} - \frac{1}{3}(x+1)^3\right]_{-1}^{0} = \boxed{\frac{1}{3}}.\end{aligned}$$

Note: If your drawing was inaccurate, you could have found the intersections algebraically: substitute $x + 1 = y^2$ into $y = (x+1)^2$ to get $y = y^4$, or $y(y-1)(y^2+y+1) = 0$. The real solutions give $(-1, 0)$ and $(0, 1)$.

Problem 1: Answer Key

(a)

Surface	Classification
(i)	Elliptic paraboloid
(ii)	Ellipsoid
(iii)	Sphere
(iv)	Hyperboloid of one sheet
(v)	None of the above (parabolic cylinder)
(vi)	None of the above (cone)

For (iii), we **complete the square [Video Reference 12.1 – Equations in 3d – 8:45]**:

$$(x^2 - 2x) + (y^2 + 4y) + (z^2 - 6z) = 2.$$

For each pair, add the square of half the linear coefficient: $(-2/2)^2 = 1$, $(4/2)^2 = 4$, and $(-6/2)^2 = 9$. Add the same amounts to the right-hand side:

$$(x^2 - 2x + 1) + (y^2 + 4y + 4) + (z^2 - 6z + 9) = 2 + 1 + 4 + 9,$$
$$(x - 1)^2 + (y + 2)^2 + (z - 3)^2 = 16.$$

The center is $(1, -2, 3)$ and the radius is $\sqrt{16} = 4$.

(b)

Function	Plot	Identifying feature
(i)	B	A single radial peak, decreasing with distance
(ii)	D	Downward paraboloid with vertex $(0, 0, 1)$
(iii)	E	A circular ridge of height 1 at $x^2 + y^2 = 2$
(iv)	F	Concentric radial oscillations
(v)	C	Alternating peaks and troughs in a grid
(vi)	A	Parallel waves along lines $x + y =$ constant

Problem 2: Answer Key

(a) $\boxed{T,\ U}$. At the local maximum T, the gradient is zero. At U, two level-curve branches with distinct tangent directions cross; the gradient must be perpendicular to both, so it is zero. **[Video Reference 14.6 – Properties of the Gradient Vector]**

(b) $\boxed{U}$. Near U, values are positive along the x-axis and negative along the y-axis, while $f(U) = 0$.

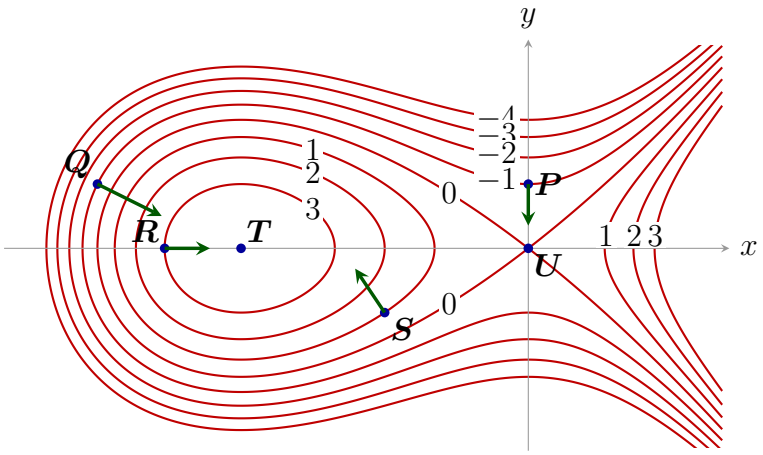
(c) $\boxed{S}$. Moving upward from S crosses contours toward larger values. At P and Q, moving upward leads toward smaller values. At R, the contour has a vertical tangent, so $f_y = 0$; the gradient vanishes at T and U.

(d) $\boxed{S}$. The directional derivative in the direction $\langle 0, -1\rangle$ is

$$\nabla f \cdot \langle 0, -1\rangle = -f_y,$$

so it is negative exactly where f_y is positive. **[Video Reference 14.6 – Directional Derivatives]**

(e) The gradients are perpendicular to the contours and point toward increasing values: downward at P, right and slightly downward at Q, right at R, and left and upward at S. At T and U, the gradient is zero. The largest magnitude is at $\boxed{Q}$, where the equally spaced contour values have the smallest perpendicular separation. **[Video Reference 14.6 – Properties of the Gradient Vector]**



Problem 3: Answer Key

(a)

Field	Plot	Identifying feature
(i)	E	Counterclockwise rotation
(ii)	C	Horizontal component vanishes on $x = y^2$
(iii)	B	Parallel to $\langle 1, 1\rangle$; zero on $x + y = 0$
(iv)	D	The components interchange x and y

[Video Reference 16.1 – Vector Fields]

(b)

Function	Gradient	Plot
(i)	$\langle x, y\rangle$	F
(ii)	$\langle y, x\rangle$	D
(iii)	$\langle 2(x + y), 2(x + y)\rangle$	B
(iv)	$\langle 2x + y, x\rangle$	A

[Video Reference 16.1 – Gradient Vector Fields]

Plots B and D are used in both parts. In plot B, the gradient in part (b) is twice the vector field in part (a), so its directions agree.

(c) Draw $x^2 + y^2 = 1$ with counterclockwise arrows on every plot.

Plot	Judgment from the picture
A	Hard to tell
B	Zero (opposite contributions cancel)
C	Hard to tell
D	Zero (opposite contributions cancel)
E	Positive (the field follows the circle)
F	Zero (the field is perpendicular to the circle)

For verification, A is the gradient of $x(x + y)$, so its closed-curve integral is zero. In C, parameterizing by $\mathbf{r}(t) = \langle \cos t, \sin t\rangle$, $0 \le t \le 2\pi$, gives

$$\langle \cos t - \sin^2 t, \sin t\rangle \cdot \langle -\sin t, \cos t\rangle = \sin^3 t,$$

whose integral over a full period is zero. Thus C also has zero integral, even though the cancellation is not obvious from the picture.

Problem 4: Answer Key

(a) $\boxed{\text{(i) B, (ii) C, (iii) A, (iv) E, (v) F, (vi) D}}$.

[Video Reference 15.2 – Type 1 and Type 2 Regions]

[Video Reference 15.3 – Double Integrals in Polar Coordinates]

(b) (i) $\displaystyle\int_0^{\pi/4}\int_0^{2} r\,dr\,d\theta$ (ii) $\displaystyle\int_0^{1}\int_y^{\sqrt{y}} dx\,dy$

(iii) $\displaystyle\int_0^{1}\int_{1-x}^{1} dy\,dx$ (iv) $\displaystyle\int_0^{\pi}\int_1^{2} r\,dr\,d\theta$

(v) $\displaystyle\int_0^{1}\int_{1-x^3}^{1} dy\,dx$ (vi) $\displaystyle\int_0^{1}\int_{\sqrt[3]{y}}^{1} dx\,dy$

For (iv), the region is the upper half-annulus between radii 1 and 2. Both polar integrals include the area factor r.